

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

VRAESTEL

Kopiereg voorbehou

Semester Test 1

QUESTION PAPER

Copyright reserved

Module EBN111

Elektrisiteit en Elektronika

Maart 2018

Module EBN111

Electricity and Electronics

March 2018



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

MEMO

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 min + 10 min to complete OCR form <i>90 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		18 + 2 Answer sheets	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. **Alle vrae moet op die ANTWOORDBLAD en op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
All questions must be answered on the ANSWER SHEET and on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

STEPS TO FOLLOW

Do step 0 before the start of the session.

Do step 1 in the first 90 minutes of the test.

Do step 2 within the following 10 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 90 min):

Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET at the end of the QUESTION PAPER. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (90 min to 100 min):

Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN111S1**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil with HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

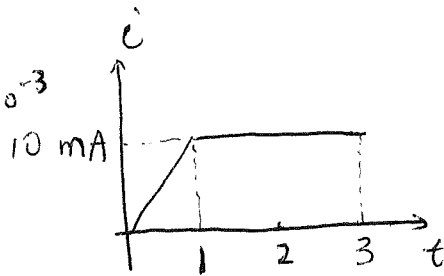
Examples

01	–	1	.	4	1					μ	A						
02	3	4	0							V							
03	–	1	5	4	.	9				k	Ω						
04	1	.	2							r	a	d	÷	s			
05	5	0	.	2	5	8				m	A						
06	1	5	5														

SECTION / AFDELING 1 [18]

Questions 1-2 [2]		Vraag 1-2 [2]	
1) The current flowing through an element increases linearly from 0 mA at $t = 0$ ms to 10 mA at $t = 1$ ms. The current stays constant from $t = 1$ ms to $t = 3$ ms at 10 mA. Calculate the total charge Q through the element from $t = 0$ ms to $t = 3$ ms.		1) Die stroom deur 'n element neem linêr toe vanaf 0 mA tot 10 mA tussen $t = 0$ ms en $t = 1$ ms. Die stroom bly konstant by 10 mA tussen $t = 1$ ms en $t = 3$ ms. Bereken die totale lading Q wat deur die element vloei tussen $t = 0$ ms en $t = 3$ ms.	
2) If a constant current of 2 A is applied to an element, and the element consumes 60 kJ energy over 1 min, what is the voltage V across the element?		2) Indien 'n konstante stroom van 2 A deur 'n element vloei, en die element verbruik 60 kJ se energie in 1 min, wat is die spanning V oor die element?	
01	Q	01	Q
02	V	02	V

$$\begin{aligned}
 01: Q &= \frac{1}{2} \times 10 \times 10^{-3} \times 10^{-3} + 2 \times 10 \times 10^{-3} \times 10^{-3} \\
 &= 25 \times 10^{-6} \text{ C} \\
 &= 25 \mu\text{C}
 \end{aligned}$$



$$02: 3.6 \times 2 \text{ V} \cdot \frac{1}{60} = 60$$

$$3.6 \times 2 \text{ V} = 3600$$

$$\Rightarrow V = 500 \text{ V}$$

Questions 3-4 [4]		Vraag 3-4 [4]	
If $R = 2\ \Omega$ and $i = 2\text{ A}$, find I_s for the independent current source, and the power P related to the dependent current source.		Indien $R = 2\ \Omega$ en $i = 2\text{ A}$, bepaal I_s vir die onafhanklike stroombron, en die drywing P geassosieer met die afhanklike stroombron.	
03	I_s	03	I_s
04	P	04	P

03: $I_s - 2i = 6 + i \Rightarrow I_s = 6 + 3i = 12\text{ A}$

04: $-10 - 8 + 2 \times 6 + V = 0$

$\Rightarrow V = 6\text{ V}$

$2i = 4\text{ A}$

$P = -6 \times 4 = -24\text{ W}.$

Question 5 [2]		Vraag 5 [2]	
05	Find R if $R_{eq} = 133 \Omega$.	05	Bepaal R indien $R_{eq} = 133 \Omega$.

$$R_{eq} = 25 + 100 + (30 + R) \parallel 10 = 133$$

$$\Rightarrow (30 + R) \parallel 10 = 8$$

$$\Rightarrow R = 10 \Omega$$

Question 6 [2]		Vraag 6 [2]	
06	Determine R_{ab} .	06	Bepaal R_{ab} .

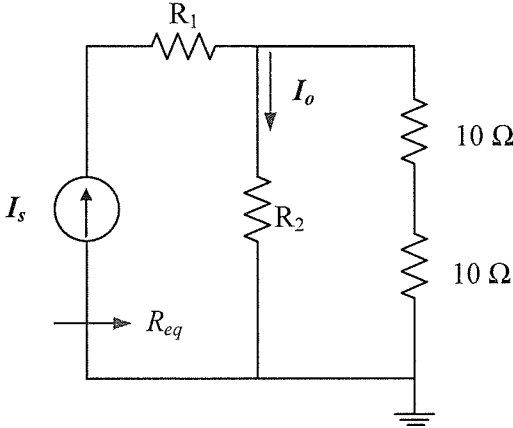
$$R = 30 \Omega.$$

$$30 \parallel 20 = \frac{30 \times 20}{50} = 12 \Omega$$

$$R_{ab} = 12 + 12 \parallel 24$$

$$= 12 + \frac{12 \times 24}{36} = 20 \Omega$$

Questions 7-8 [4]		Vraag 7-8 [4]	
1) Find R_1 if $R_2 = 5 \Omega$ and $R_{eq} = 20 \Omega$		1) Bepaal R_1 indien $R_2 = 5 \Omega$ en $R_{eq} = 20 \Omega$.	
2) Find R_2 if $\frac{I_o}{I_s} = \frac{1}{2}$.		2) Bepaal R_2 indien $\frac{I_o}{I_s} = \frac{1}{2}$.	
07	R_1	07	R_1
08	R_2	08	R_2



(Note, R_{eq} includes R_1 . R_{eq} is the total resistance of the circuit.
 Let op, R_{eq} sluit R_1 in. R_{eq} is die totale weerstand van die kringbaan.)

$$1) R_2 = 5, R_2 || 20 = \frac{5 \times 20}{5 + 20} = 4 \Omega$$

$$R_{eq} = R_1 + 4 = 20 \Rightarrow R_1 = 16 \Omega$$

$$2) \text{ if } I_o / I_s = \frac{1}{2} \Rightarrow R_2 = 20 \Omega$$

Question 9-10 [4]		Vraag 9-10 [4]	
If $V_s = 8\text{ V}$, determine i_o and V_{ab} .		Indien $V_s = 8\text{ V}$, bepaal i_o en V_{ab} .	
09	i_o	09	i_o
10	V_{ab}	10	V_{ab}

$$V_a = 8 \cdot \frac{6}{8} = 6\text{ V.}$$

$$V_b = 8 \cdot \frac{3}{8} = 3\text{ V}$$

$$\Rightarrow V_{ab} = V_a - V_b = 3\text{ V;}$$

$$R_T = 80 \parallel 80 = 40\Omega.$$

$$\Rightarrow i_o = \frac{8}{40} = 0.2\text{ A.}$$

SECTION/AFDELING 2 [20]

Questions 11-12 [4]		Vraag 11-12 [4]	
Determine θ if $v_2 = 1$. $\begin{bmatrix} 3 & \theta \\ 12 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \theta \\ -4\theta \end{bmatrix}$		11	Bepaal θ indien $v_2 = 1$. $\begin{bmatrix} 3 & \theta \\ 12 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} \theta \\ -4\theta \end{bmatrix}$
Determine σ if $i_1 = 1$. $\begin{bmatrix} 2 & 1 & 4 \\ 0 & \sigma & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix}$		12	Bepaal σ indien $i_1 = 1$. $\begin{bmatrix} 2 & 1 & 4 \\ 0 & \sigma & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ -2 \end{bmatrix}$
11	θ	11	θ
12	σ	12	σ

$$v_2 = \frac{\begin{vmatrix} 3 & \theta \\ 12 & -4\theta \end{vmatrix}}{\begin{vmatrix} 3 & \theta \\ 12 & -4 \end{vmatrix}} = 1 \Rightarrow \theta = 1$$

$$i_1 = \frac{\begin{vmatrix} 1 & 1 & 4 \\ 5 & \sigma & 1 \\ -2 & 4 & 0 \end{vmatrix}}{\begin{vmatrix} 2 & 1 & 4 \\ 0 & \sigma & 1 \\ 2 & 4 & 0 \end{vmatrix}} = 1$$

$$\Rightarrow \sigma = -5$$

Question 13-16 [4]				Vraag 13-16 [4]			
Use mesh analysis to set up two equations of the following form:				Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:			
$ai_1 + bi_3 = 4$				$ai_1 + bi_3 = 4$			
$ci_1 + di_3 = 18$				$ci_1 + di_3 = 18$			
Calculate the unknown coefficients a, b, c , and d .				Bepaal die onbekende koëffisiënte a, b, c , en d .			
13	a	14	b	13	a	14	b
15	c	16	d	15	c	16	d

$$i_1 - i_3 = 4$$

$$i_2 = 2$$

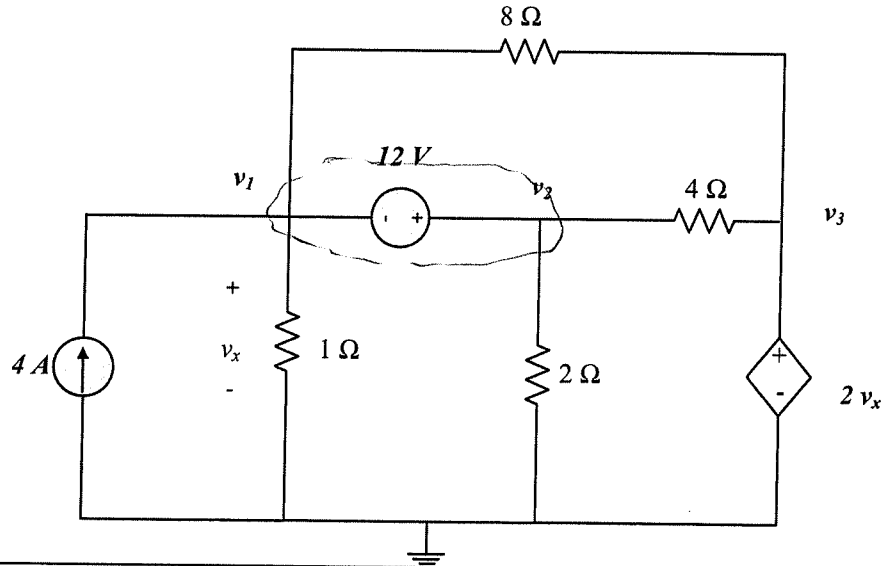
$$3i_1 + 12i_3 + 10(i_3 - 2) + 2 = 0$$

$$3i_1 + 22i_3 = 18$$

$$a = 1, \quad b = -1$$

$$c = 3, \quad d = 22$$

Question 17-20 [4]				Vraag 17-20 [4]			
Use nodal analysis to set up two equations of the following form:				Gebruik nodeanalyse om twee vergelykings in die volgende vorm op te stel:			
$eV_1 + fV_2 = -12$ $gV_1 + hV_2 = 32$				$eV_1 + fV_2 = -12$ $gV_1 + hV_2 = 32$			
Calculate the unknown coefficients e, f, g , and h .				Bepaal die onbekende koëffisiënte e, f, g , en h .			
17	e	18	f	17	e	18	f
19	g	20	h	19	g	20	h



$$V_1 + 12 = V_2 \Rightarrow V_1 - V_2 = -12$$

$$-4 + \frac{V_1}{1} + \frac{V_2}{2} + \frac{V_2 - V_3}{4} + \frac{V_1 - V_3}{8} = 0$$

$$V_3 = 2V_x = 2V_1$$

$$-32 + 8V_1 + 4V_2 + 2V_2 - 2V_3 + V_1 - V_3 = 0$$

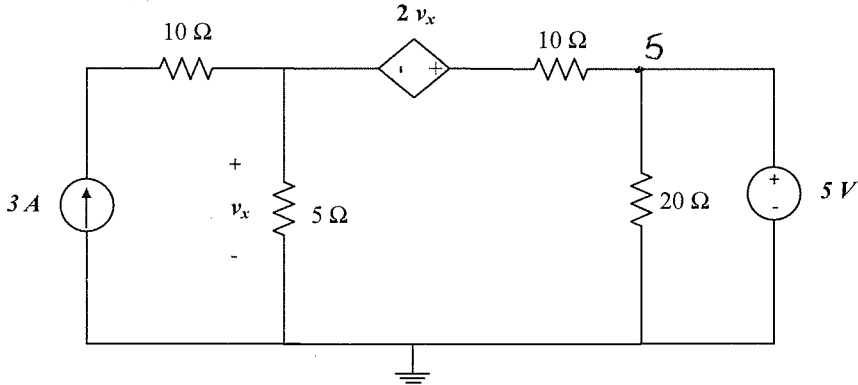
$$9V_1 + 6V_2 - 3V_3 = 32$$

$$3V_1 + 6V_2 = 32$$

$$\Rightarrow e = 1, f = -1, g = 3, h = 6$$

Question 21-22 [2]		Vraag 21-22 [2]	
Use inspection to set up the matrix equation in terms of the mesh currents. What are the values of p and q ?		Gebruik inspeksie om 'n matriks vergelyking in terme van die lusstrome neer te skryf. Wat is die waardes van p en q ?	
$\begin{bmatrix} 5 & -4 \\ -4 & p \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} -10 \text{ V} \\ q \end{bmatrix}$		$\begin{bmatrix} 5 & -4 \\ -4 & p \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} -10 \text{ V} \\ q \end{bmatrix}$	
21	p	21	p
22	q	22	q

~~$p = 8$~~ $p = 12$
 $q = -2 \text{ V}$

Question 23 [2]		Vraag 23 [2]	
Calculate v_x using either nodal or mesh analysis.		Bepaal v_x deur gebruik te maak van node- of lus-analise.	
23	v_x	23	v_x
			

$$-3 + \frac{V_x}{5} + \frac{V_x + 2V_x - 5}{10} = 0$$

$$-30 + 2V_x + 3V_x - 5 = 0$$

$$\Rightarrow 5V_x = 35$$

$$V_x = 7V$$

Questions 24-25 [4]		Vraag 24-25 [4]	
Given $\beta = 99$ and $V_{BE} = 0.7$ V, 1) Determine V_{CE} if $I_B = 25 \mu\text{A}$ and $R_E = 0 \Omega$. 2) Determine i_B if $V_S = 2$ V and $R_E = 300 \Omega$.		Gegee $\beta = 99$ en $V_{BE} = 0.7$ V, 1) Bepaal V_{CE} indien $I_B = 25 \mu\text{A}$ en $R_E = 0 \Omega$. 2) Bepaal i_B indien $V_S = 2$ V en $R_E = 300 \Omega$.	
24	V_{CE}	24	V_{CE}
25	i_B	25	i_B

$$1) \quad \beta i_B + \frac{V_{CE}}{1k} + \frac{V_{CE}-8}{2k} = 0$$

$$\Rightarrow 2k \cdot \beta i_B + 2V_{CE} + V_{CE} - 8 = 0$$

$$\Rightarrow 3V_{CE} = 8 - 2 \times 10^3 \times 99 \times 25 \times 10^{-6}$$

$$\Rightarrow \underline{V_{CE} = 1.0167 \text{ V}}$$

$$2) \quad -2 + 100k \cdot i_B + 0.7 + (\beta + 1)i_B \times 300 = 0$$

$$100 \times 10^3 i_B + 100 \times 300 i_B = 1.3$$

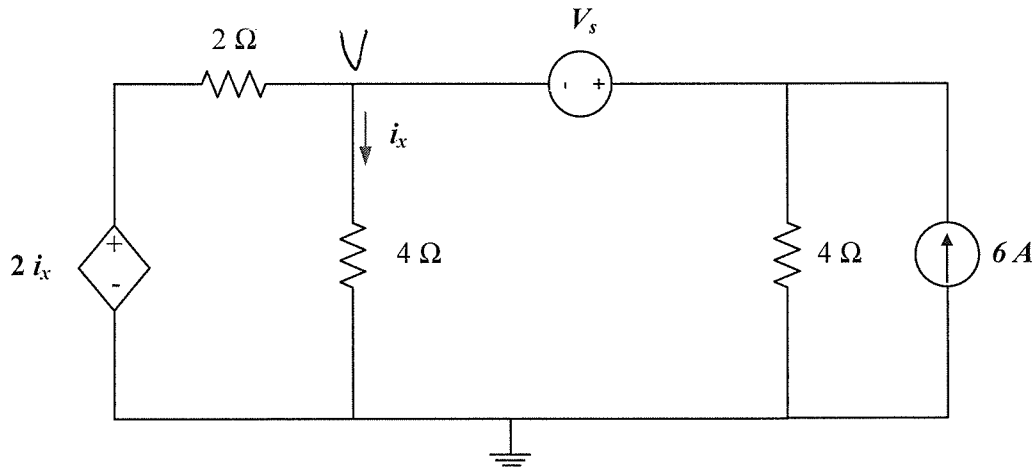
$$(100 \times 10^3 + 30 \times 10^3) i_B = 1.3$$

$$\Rightarrow 130 i_B = 1.3 \times 10^{-3}$$

$$\Rightarrow \underline{i_B = 10 \mu\text{A}}$$

SECTION / AFDELING 3 [8]

Questions 26-27 [4]		Vraag 26-27 [4]	
1) Find the exclusive contribution of the 6 A current source to the current i_x .		1) Bepaal die eksklusiewe bydrae van die 6 A stroombron tot die stroom i_x .	
2) Find V_s if the voltage source V_s has an exclusive contribution of 2 A to the current i_x .		2) Bepaal V_s indien die spanningsbron V_s 'n eksklusiewe bydrae van 2 A tot die stroom i_x het.	
26	$i_{x(6A)}$	26	$i_{x(6A)}$
27	V_s	27	V_s



1) Let $V_s = 0$;

$$\frac{V - 2i_x}{2} + \frac{V}{4} + \frac{V}{4} - 6 = 0$$

$$\frac{V - 2i_x}{2} + \frac{V}{2} - 6 = 0 \Rightarrow V - i_x = 6$$

$$V - \frac{V}{4} = 6 \Rightarrow V = 8V \Rightarrow \underline{i_x = \frac{8}{4} = 2A} \rightarrow$$

2) $\frac{V}{4} = i_x \Rightarrow V = 4i_x$, Let 6A be switched off.

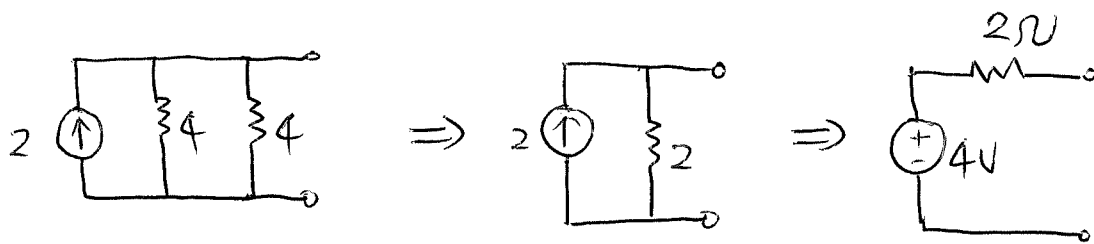
$$i_x = 2A \Rightarrow V = 8, \quad 2i_x = 4$$

$$\frac{8 - 4}{2} + 2 + \frac{8 + V_s}{4} = 0 \Rightarrow \underline{V_s = -24V} \rightarrow$$

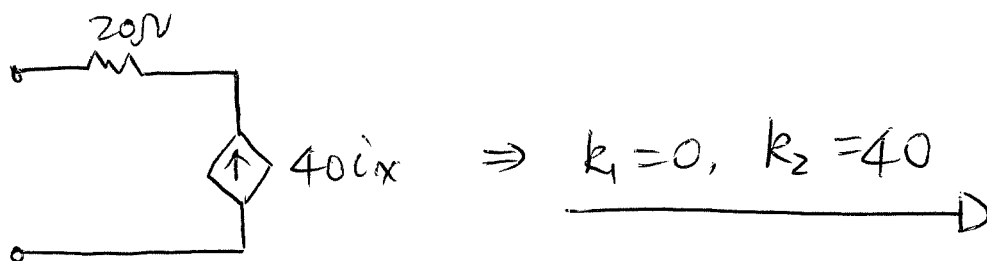
Question 28-31 [4]		Vraag 28-31 [4]	
Circuit (a) can be transformed into circuit (b) Find k_1 , k_2 , R , and i_x .		Kringbaan (a) kan na kringbaan (b) getransformeer word. Bepaal k_1 , k_2 , R , en i_x .	
28	k_1	28	k_1
29	k_2	29	k_2
30	R	30	R
31	V_s	31	i_x

(a)

(b)



$$V_s = 4V, R = 2\Omega;$$



PRACTICE PAGE / OEFENBLAD

Assessment ID

2	0	1	8	E	B	N	1	1	1	S	1		
---	---	---	---	---	---	---	---	---	---	---	---	--	--

Answer

Unit

SECTION / AFDELING 1		
01	$Q = 25$	μC
02	$V = 500$	V
03	$I_s = 12$	A
04	$P = -24$	W
05	$R = 10$	Ω
06	$R_{ab} = 20$	Ω
07	$R_1 = 16$	Ω
08	$R_2 = 20$	Ω
09	$i_o = 0.2$	A
10	$V_{ab} = 3$	V
SECTION / AFDELING 2		
11	$\theta = 1$	No units required
12	$\sigma = -5$	
13	$a = 1$	
14	$b = -1$	
15	$c = 3$	
16	$d = 22$	
17	$e = 1$	
18	$f = -1$	
19	$g = 3$	
20	$h = 6$	
21	$p = 8$ $p=12$	
22	$q = -2$	V
23	$v_x = 7$	V
24	$V_{CE} = 1.0167$	V
25	$i_B = 10$	μA

SECTION / AFDELING 3		
26	$i_{x(6A)} = 2$	A
27	$V_s = -24$	✓
28	$k_1 = 0$	No units required
29	$k_2 = 40$	
30	$R = 2$	Ω
31	$V_s = 4$	V